Learnable Representative Coefficient Image Denoiser for Hyperspectral Image

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***Abstract*— Fully characterizing the spatial-spectral priors of hyperspectral images (HSIs) is crucial for HSI denoising tasks. Recently, HSI denoising models based on representative coefficient images (RCIs) under the spectral low-rank decomposition framework have garnered significant attention due to their clever utilization of spatial-spectral information in HSI at a low cost. However, current methods either employ handcrafted classical denoisers or off-the-shelf deep denoisers to denoise RCIs, failing to fully capture the structural information of RCIs. In this article, we propose a specific optimization framework for learning an RCI denoiser under the low-rank decomposition framework for the first time. Since low-rank decomposition can characterize the global low-rank property of HSI, our RCI denoiser only needs to learn the spatial prior of RCIs. Consequently, our optimization framework is inclined to learn a more powerful RCI denoiser. However, learning an RCI denoiser is not an easy task, primarily due to the lack of paired clean-noisy RCI data. To address this issue, we employ parametric techniques to represent the to-be-restored HSI as a function of RCI denoiser network parameters. In this way, the parameters of the RCI denoiser can thus be updated using noisy-clean HSI pairs. Furthermore, we adopt residual learning and Gaussian whitening (GW) techniques to enhance the RCI denoiser's denoising ability for HSIs with various noise levels and different rank settings. Extensive experiments demonstrate that our method can achieve significant improvements in both denoising effectiveness and speed compared to state-of-the-art methods. The code of our algorithm is released at https://github.com/andrew-pengjj/RCILD.git.**

***Index Terms*— Gaussian whitening (GW), learnable deep RCI denoiser, representative coefficient images (RCIs), residual learning.**

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I. Introduction

HYPERSPECTRAL image (HSI) is a high-dimensional data cube containing a series of 2-D images collected from sensors over various bands that provide rich spectral information, and thus has been widely used in many fields, for example, classification [1], super-resolution [2], compressive sensing [3], unmixing [4], and mineral exploration [5]. With the existence of the instrumental error, HSI is often corrupted by Gaussian noise, which significantly hinders its subsequent applications. It is thus an ongoing hot topic to explore efficient HSI denoising algorithms.

For HSI denoising, how to make good use of its structural information priors is the key to success. The HSI priors can be roughly divided into two categories, that is, the basic spectral low-rankness [6], [7], [8], [9] and other rich spatial priors [10], [11], [12], [13], [14]. Around the characterization of prior information in HSI, two main categories of methods have emerged: model-driven methods [7], [9], [11], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], data-driven methods [21], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]. These two categories of methods will be elaborately introduced in Section II.

Among all these denoising methods, a recently popular category based on subspace denoising has garnered widespread attention [12], [20], [24], [31], [32], [37]. As shown in Fig. 1, after performing low-rank decomposition along the spectral dimension of the original noisy image $\mathcal{Y} \in \mathbb{R}^{M\times N\times B}$, one can obtain a tensor $\mathcal{U} \in \mathbb{R}^{M\times N\times R}$ consisting of a series of representative coefficient images (RCIs) and an orthogonal basis matrix $\mathbf{V} \in \mathbb{R}^{R\times B}$. Compared to the original noisy image $\mathcal{Y} \in \mathbb{R}^{M\times N\times B}$, each slice of the RCIs exhibits a texture structure that closely resembles the clean image $\mathcal{X} \in \mathbb{R}^{M\times N\times B}$. This observation indicates that the RCIs retain the essential information from the original HSI data and are more robust to noise [24]. Additionally, the number of slices (i.e., R) in the RCIs is significantly lower than that in the original data due to its low-rank nature. Therefore, by leveraging this property, we can decompose large-scale data denoising tasks into

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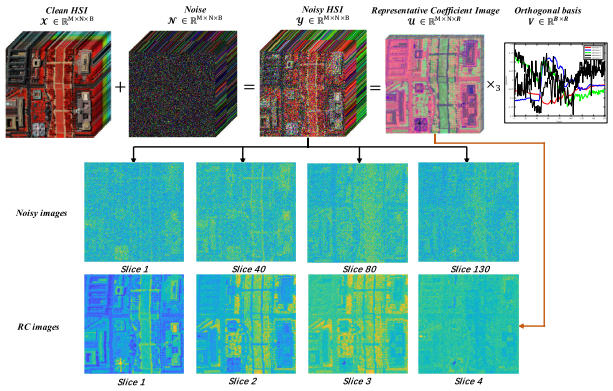


Fig. 1. Comparison between the noisy HSI images and RCIs under the spectral low-rank decomposition framework.

two smaller-sized data restoration problems, thus achieving an acceleration effect. However, existing subspace denoising methods either rely on manually designing a regularization term [12], [20], [24] or directly applying a neural network pre-trained on natural images as the denoiser for RCIs [31], [32]. Such limitation results in inadequate exploration of prior knowledge for the RCIs of HSIs. If we can obtain a better RCI denoiser, there is still potential for further performance improvement in subspace denoising methods.

To obtain a more effective RCI denoiser for HSIs, a natural idea is to train an RCI denoiser specifically tailored for HSIs. However, training a denoiser for the RCIs is not a simple task. The difficulty arises from the fact that the noise is added to the original data, and there is no one-to-one correspondence between the RCIs obtained from the low-rank decomposition of the original data and those obtained from the low-rank decomposition of the noise maps, that is, each RCI obtained from one noisy HSI does not match exactly with that from the clean HSI. As a result, it is challenging to construct paired clean-noisy training examples for the RCIs. This is also the reason why there is currently no dedicated work on training deep denoisers specifically for the RCIs. Constructing clean-noisy RCI data pairs poses a challenge. However, we can construct lots of clean-noisy HSI data pairs. By using parametric techniques, we can represent the recovered HSI data through the network parameters of the RCI deep denoiser within a low-rank decomposition framework, then we can use paired clean-noisy HSI data to update the network parameters of the RCI denoiser. To the best of our knowledge, this should be the first work to learn a deep denoiser for RCIs. Moreover, we observed considerable scale discrepancies between different RCIs when constrained by a low-rank orthogonal decomposition (i.e., when the matrix $\mathbf{V}$ procured in Fig. 1 is orthogonal). Predicated on this characteristic, we adopt the classic residual learning strategy in our learning framework. Furthermore, to augment the model's generalization capacity to handle non-i.i.d noise, we adopted Gaussian whitening (GW) techniques, which have demonstrated remarkable efficacy in HSI denoising [20], [21], [32]. Fig. 2 and Table I present a comparison of the performance of current subspace learning-based methods. In Table I, Adhyde [31] uses a pre-trained network named FFD-net [38], while our

TABLE I

QUANTITATIVE COMPARISONS ON CAVE AND DC MALL DATASETS UNDER I.I.D GAUSSIAN NOISE WITH THE VARIANCE $\sigma = 0.40$. HERE, OUR METHOD CHOOSES DNCNN AS THE RCI DENOISER, THE REASON BEHIND WHICH CAN BE SEEN IN SECTION III-D

Dataset	Metric	Noisy	Fast-HyDe [20]	NG-meet [37]	Ad-hyde [31]	Ad-hyde [31]	Ours
DC mall	PSNR	7.96	29.98	30.10	29.57	29.74	**31.16**
	SSIM	0.088	0.910	0.906	0.907	0.905	**0.927**
ICVL	PSNR	7.95	32.64	34.56	32.15	32.42	**35.78**
	SSIM	0.119	0.871	0.897	0.866	0.868	**0.958**
RCI denoiser		/	BM3D	WNNM	BM3D	FFD-net	DnCNN
Is learnable?		/	No	No	No	No	Yes

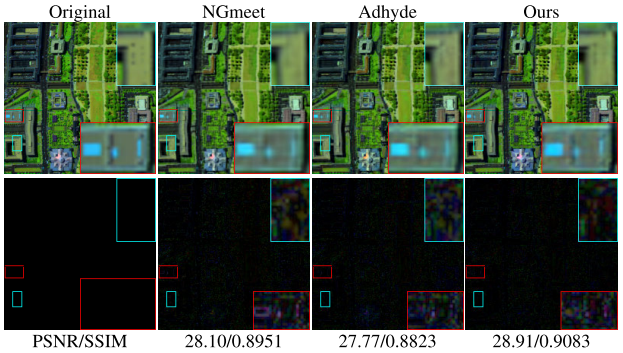


Fig. 2. Recovered false images (R: 100, G: 50, B: 1) and its error maps on DC mall dataset under Gaussian noise $\mathcal{N}(0, 0.4^2)$.

method chooses DnCNN [39] network as the RCI denoiser and uses the paired clean-noise HSI data to learn the parameter of DnCNN. From Fig. 2 and Table I, it is evident that our proposed learning-based approach significantly outperforms previous methods, highlighting the importance of learning-based techniques.

In summary, this article makes the following three contributions.

1) We use pairs of noisy clean HSI images to update the parameters in the RCI denoiser via parametrically representing the repaired HSI data under a low-rank decomposition framework, thus avoiding the need for clean RCIs, which makes us learn a first specialized deep RCI denoiser.
2) We train a unified deep denoiser for all RCIs through the residual learning strategy and GW techniques, which makes it suitable for HSI with various noise levels and ranks. Thus the proposed method is very user-friendly.
3) Extensive experiments demonstrate the superiority of the proposed method both in denoising effect and speed. Specifically, it achieves state-of-the-art performance within only one second for most test data.

The rest of this article is organized as follows: Section II gives some related works. In Section III, we give the detail description about our optimization process about learning a deep RCI denoiser. Experimental results are shown in Section IV. Finally, conclusions are drawn in Section V.

II. RELATED WORK

In this section, we briefly review the existing model-based methods and some data-based methods.

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A. Model-Driven Methods

1) HSI Denoising in Original Space: Model-driven approaches focus on how to propose proper regularizations to finely characterize priors on HSI data. Among all priors, the spectral low-rank property is the most important one for HSI, which is modeled by matrix/tensor nuclear norm or low-rank matrix decomposition/low-rank tensor decomposition (LRMD/LRTD) [11], [13], [21], [40], [41], [42], [43]. Typical methods include RPCA [15], WNNM [7], [44], TRPCA [16], [45], HyRes [46], and many others [17], [18], [19], [47]. However, only utilizing the spectral low-rank property is not enough to obtain an optimal denoising performance since it ignores the spatial priors. The non-local similarity and local smoothness are the two most typical types of spatial priors and these two priors are often integrated into the LRMD/LRTD framework. The non-local spatial similarity prior refers to the fact that there are always similar patterns in image spatial dimensions, thus this prior is often integrated into low-rank matrix/tensor approximation frameworks to remove Gaussian noise. The classical denoising methods based on non-local similarity include BM4D [48], KBR [9], LLRT [14], and many others [49], [50]. The local smoothness prior is based on the fact that similar objects are often distributed in a local area with high probability. Besides, the adjacent bands of images are usually collected with similar sensor parameter settings, which results in similar values. Such local smoothness prior can be encoded by using total variation (TV) regularization on the spatial and spectral domain of the HSI [13], [51]. The classical denoising methods based on TV regularization include LRTV [11], LRTDTV [13], LRTDGS [23], E3DTV [52], CTV-RPCA [51], and many others [53], [54].

Model-based methods are built upon statistical prior modeling, which offers excellent interpretability and generalization. However, due to the relatively large dimensionality of the original data, computing regularizers in the original space can be time-consuming. Moreover, manually designed regularizers often fail to precisely capture the prior structures of the data.

2) HSI Denoising in RCI: To address the efficiency issue, some methods based on subspace prior modeling have been proposed. The main idea is that the spatial prior of the original data can be transferred to its RCIs under low-rank decomposition framework. Low-rank decomposition ensures the spectral low-rank property of HSI. Therefore, by effectively characterizing the prior information of RCIs, we can simultaneously leverage both the spectral low-rank property and the spatial prior of HSI. Consequently, the large-sized data restoration task can be transformed into two small-sized matrices, facilitating accelerated algorithms.

Many works have shown that the RCIs exhibit similar structures to the original HSI, possessing properties such as local smoothness, non-local self-similarity, and other priors. For instance, NGmeet [12], [37] and FastHyDe [20] achieve fast and high-precision denoising by leveraging the non-local self-similarity of RCIs. RCTV [24] and LRTFDFR [22] achieve good results in removing mixed noise by exploiting the local smoothness of RCIs.

However, the denoisers for RCIs in the above-mentioned models are manually designed, making it arduous to precisely characterize the priors of RCIs. To address this issue, some research has started incorporating deep learning into the subspace denoising framework, such as Adhyde [31] and FastHyMix [32]. However, these methods simply adopt a pre-trained network, such as FFD-net [38], which is trained on natural images, to serve as the denoiser for RCIs. Due to the significant differences in the properties between natural images and RCIs, and the distinct noise distributions on the original images compared to RCIs, directly using a pre-trained network cannot fully capture the priors of RCIs.

B. Data-Driven Methods

Unlike model-driven methods, data-driven methods directly learn a denoising mapping from a large amount of data. So far, many complex network architectures [25], [26], [27], [28], [29], [30], [33], [34] have been proposed. For example, HSICNN [26] captures the low-rank property of an HSI by inputting a picture and several pictures before and after it. 3D-ADNet [27] adopts 3-D convolution instead of 2-D convolution together with self-attention to extract HSI features. QRNN3D [28] utilizes 3-D convolution and quasi-recurrent pooling functions to characterize the structural spatial-spectral correlation and global correlation along the spectral dimension. GRN-net [29] adopts two reasoning modules to carefully extract both global and local spatial-spectral features for mixed noise removal. T3SC [30] proposes a model-guided spectral-spatial network by unfolding the iterative algorithm to solve the sparse model. The aforementioned data-driven methods are powerful in exploring image spatial information but may not be sufficient to capture the spectral low-rank structure of HSIs. However, modeling the spectral low-rank structure is extremely straightforward for low-rank decomposition methods.

In addition to supervised learning, there are also unsupervised deep learning models in the field of HSI denoising, such as DHP [55] and S2DIP [56]. These methods are mainly based on the inherent capability of convolutional neural networks to encode image priors. However, these methods face a challenge as they operate in an unsupervised learning framework. Without early stopping, the network might end up fitting to the noisy data, resulting in poor denoising performance. Moreover, currently, there is a lack of a suitable criterion for early stopping, making its practical application less convenient.

III. Proposed Approach

A. Problem Formulation

For an HSI $\mathcal{X} \in \mathbb{R}^{M\times N\times B}$, it can be decomposed as $\mathcal{X} = \mathcal{U} \times_3 \mathbf{V}$ using spectral low-rank decomposition, where $\mathbf{V} \in \mathbb{R}^{B\times R}$ ($R \ll B$) is the spectral subspace consisting of R orthogonal basis and $\mathcal{U} \in \mathbb{R}^{M\times N\times R}$ is the corresponding coefficients, which are also called as RCIs. Considering its corruption by additive Gaussian noise $\mathcal{E}$, the noisy HSI $\mathcal{Y}$ can then be generated by

$$\mathcal{Y} = \mathcal{U} \times_3 \mathbf{V} + \mathcal{E}. \tag{1}$$

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Next, we will illustrate that one can denoise the HSI by mining the rich spatial structures of these RCIs since we can prove that the RCI inherits abundant spatial information from the original HSI via Theorem 1.

Theorem 1: For a rank-R matrix $\mathbf{X} \in \mathbb{R}^{MN\times B}$ that has the factorization $\mathbf{X} = \mathbf{U}\mathbf{V}^{\mathrm{T}}$ with orthogonal matrix $\mathbf{V} \in \mathbb{R}^{B\times R}$, its representative coefficient matrix can be obtained by $\mathbf{U} = \mathbf{X}\mathbf{V}$, and we have.

1) If the two row vectors $\mathbf{X}(i,:)$ and $\mathbf{X}(j,:)$ of original data $\mathbf{X}$ are the same, then their corresponding coefficient vectors $\mathbf{U}(i,:)$ and $\mathbf{U}(j,:)$ are the same.
2) The distance and angle between any two-row vectors of $\mathbf{X}$ are the same as those of the corresponding two-row vectors of $\mathbf{U}$.

The proof of Theorem 1 can be easily proven [24]. In fact, many studies have demonstrated that, within the low-rank decomposition framework, more robust denoising effects can be achieved by exploring the prior structure of RCIs, such as [52], [57], [58]. Mathematically speaking, such a modeling paradigm can be formulated as follows:

$$\min_{\mathcal{U},\mathbf{V}} \mathcal{R}(\mathcal{U}) + \frac{\mu}{2}\|\mathcal{Y} - \mathcal{U}\times_3 \mathbf{V}\|_F^2 \quad \text{s.t.} \quad \mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I}, \tag{2}$$

where $\mathcal{R}(\mathcal{U})$ denotes a certain regularizer encoding the structure information on $\mathcal{U}$ and μ is a trade-off parameter. Alternating optimization strategy with iterative refinement [7], [14], [37], [44] has been verified to be an effective tool for solving (2). Following the update process in NGmeet [37], the entire iterative process of (2) can be written as follows:

$$\begin{cases} \{\overline{\mathcal{U}}_k, \mathbf{V}_k\} = \arg\min\limits_{\mathcal{U},\mathbf{V}^{\mathrm{T}}\mathbf{V}=\mathbf{I}} \|\mathcal{Y}_k - \mathcal{U}\times_3 \mathbf{V}\|_F^2, \\ \mathcal{U}_k = \arg\min\limits_{\mathcal{U}} \mathcal{R}(\mathcal{U}) + \dfrac{\mu}{2}\|\mathcal{U} - \overline{\mathcal{U}}_k\|_F^2, \\ \mathcal{X}_k = \mathcal{U}_k \times_3 \mathbf{V}_k, \\ \mathcal{Y}_{k+1} = \lambda\mathcal{X}_k + (1-\lambda)\mathcal{Y}, \end{cases} \tag{3}$$

where $0 < \lambda < 1$ is a refine parameter and $\mathcal{X}_k$ is the recovered HSI data at kth iteration. For (3), the first subproblem of $\{\overline{\mathcal{U}}_k, \mathbf{V}_k\}$ can be viewed as low-rank decomposition and can be solved by the following Lemma 1. The proof of Lemma 1 can be seen in [37] and [52]. For better clarity, we illustrate the process of (3) in Fig. 3.

Lemma 1: Let $(\mathcal{Y}_k)_{(3)} = \mathbf{U}\Sigma\mathbf{V}^{\mathrm{T}}$ be rank-R SVD of $(\mathcal{Y}_k)_{(3)}$. The solution to the first subproblem in (3) is given by the closed-form as $\mathbf{V}_k = \mathbf{V}$ and $\overline{\mathcal{U}}_k = \mathrm{fold}_3(\mathbf{U}\Sigma)$.

B. RCI Denosier

After getting $\{\overline{\mathcal{U}}_k, \mathbf{V}_k\}$, the core difference of RCI-based models falls on how to design a better RCI regularizer, that is, $\mathcal{R}(\mathcal{U})$, or design the effective methods to solve the following problem:

$$\mathcal{U}_k = \arg\min_{\mathcal{U}} \mathcal{R}(\mathcal{U}) + \frac{\mu}{2}\|\mathcal{U} - \overline{\mathcal{U}}_k\|_F^2. \tag{4}$$

Existing methods to solve (4) include WNNM [7], [44], BM3D [20], successive SVD [59], TV [24], and CNN [31], [32]. But all of them either use hand-designed denoiser or off-the-shelf deep denoiser trained on natural images, which cannot fully capture the structures of the RCIs.

To alleviate the aforementioned issue, we can view (4) as a classic proximal gradient optimization problem and embed it into a deep learning framework for training, aiming to better capture the prior information of RCI, and perform denoising. Denoting the network for solving (4) as $\Phi_k(\cdot)$ for kth iteration, the solution to (4) can be expressed as follows:

$$\mathcal{U}_k := \Phi_k(\overline{\mathcal{U}}_k, \Theta_k). \tag{5}$$

For the proximal gradient network $\Phi_k(\cdot)$, the input and output of the network are $\overline{\mathcal{U}}_k$ and $\mathcal{U}_k$, respectively, and Θ_k represents the network parameters that need to be learned.

C. Parametric Representation

However, updating the network weight Θ_k in (5) is not a simple task. The difficulty arises from the fact that the noise is added to the original HSIs, and there is no one-to-one correspondence between the RCIs obtained from the low-rank decomposition of the HSIs and those obtained from the low-rank decomposition of the noise maps, that is, each RCI obtained from one noisy HSI does not match exactly with that from the clean HSI. As a result, it is challenging to construct paired clean-noisy training examples for the RCIs. This is also the reason why there is currently no dedicated work on training deep denoisers specifically for the RCIs. To address this issue, we employed the following parametric technique. Although obtaining paired noisy-clean RCI training sets is hard, we can easily obtain paired noisy-clean HSI training sets. This inspired us to represent the denoised HSI as a function of the RCI denoising network parameters Θ_k. This technique allows us to utilize clean HSI data to update the parameters Θ_k of the RCI denoisier $\Phi_k(\cdot, \Theta_k)$. The complete process is as follows:

$$\begin{cases} \mathcal{U}_k(\Theta_k) & := \Phi_k(\overline{\mathcal{U}}_k, \Theta_k), \\ \mathcal{X}_k(\Theta_k) & = \mathcal{U}_k \times_3 \mathbf{V}_k. \end{cases} \tag{6}$$

Since $\mathcal{X}_k(\Theta_k)$ is the parametric representation of Θ_K, we can directly use the clean HSI data $\mathcal{X}_{\mathrm{gt}}$ to update the parameters of the RCI denoisier $\Phi_k(\overline{\mathcal{U}}_k, \Theta_k)$ without the clean RCIs. Therefore, the network's loss function is

$$\arg\min_{\Theta_k} \|\mathcal{X}_k(\Theta_k) - \mathcal{X}_{\mathrm{gt}}\|_F^2. \tag{7}$$

D. Structure of RCI Denosier

Here, we aim to establish a denoising network that takes a single RC image as input. Consequently, our RCI denoiser allows our model to achieve better denoising performance on HSI data with different low-rank characteristics by adjusting the number of RCIs (i.e., the rank in low-rank decomposition). To achieve this goal, we must analyze the property of RCIs. In Fig. 4, we show several RCIs and the Frobenius norm curve of all RCIs under the low-rank decomposition framework on two HSI data. From these figures, we can see that there are large differences in the scale and brightness of different RCIs. In reality, each RCI represents a principal component, and under orthogonal low-rank decomposition, the higher-ranked RCIs contain more information from the original data. Therefore, there is a particularly large difference in grayscale

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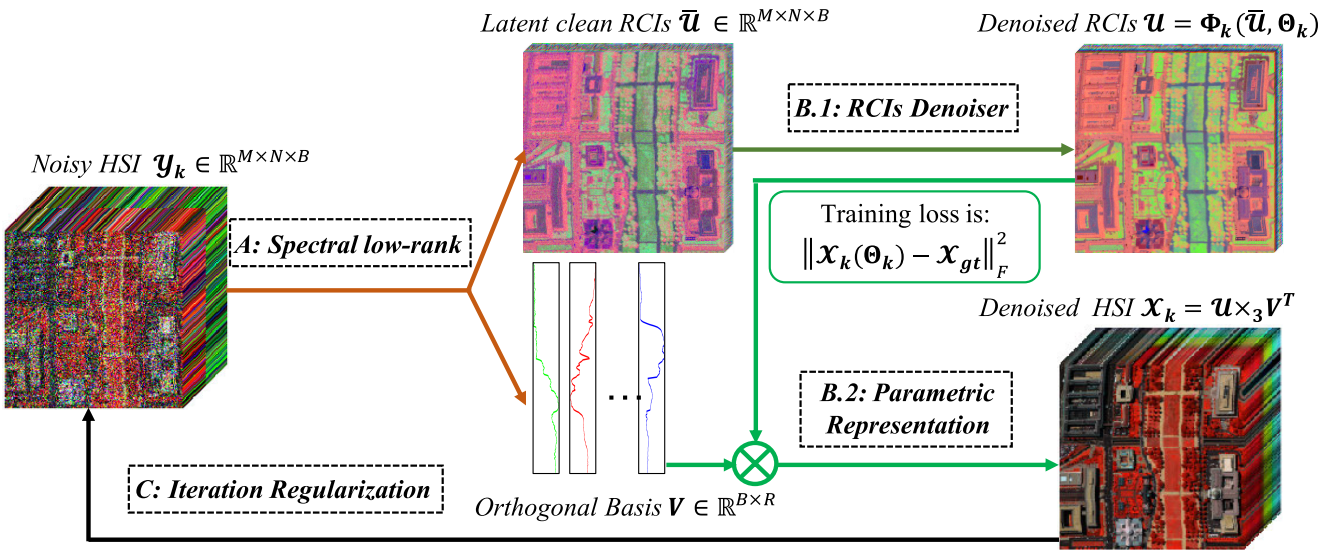


Fig. 3. Flowchart of the proposed method. It includes three stages. A: spectral low-rank decomposition. B: RCI denoising via deep network. C: iteration regularization.

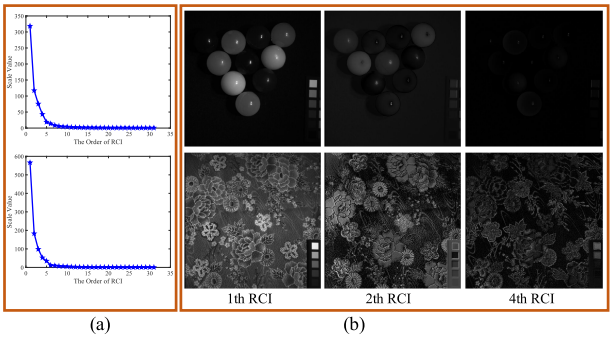


Fig. 4. (a) Scale curve of RCIs. (b) Brightness map display of some RCIs.

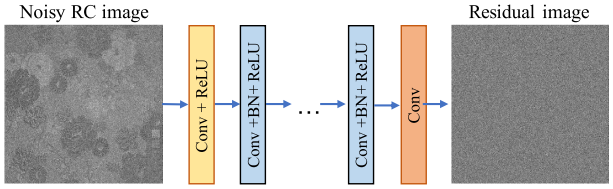


Fig. 5. Structure of DnCNN network.

size between different RCIs. This inspires us to directly learn the mapping from noisy image to its residual. DnCNN [39] is a classic residual learning-based denoising network. Its network structure includes convolution, batch normalization (BN), and ReLU, and has been proven to be very effective in the field of image denoising; the structure of DnCNN is plotted in Fig. 5.

Thus, we choose a DnCNN network as the RCI denoiser. We also provide the ablation experiments of different network structures in the experiment section. In addition, as can be seen from Fig. 4, these RCIs are non-trivial image-like matrices whose gray value distribution is completely different from that of natural images, so it is necessary for us to learn a specialized deep RCI denoiser for HSI. As we trained a unified denoiser for all the RCIs, it is crucial to ensure that the number of layers in DnCNN is neither too small to capture common patterns, nor too large, as it may result in slower training. Therefore, we chose a 15-layer DnCNN, with a header consisting of a "Conv+Relu" layer, 13 intermediate layers containing "Conv+BN+Relu" operations, and a tail consisting of a "Conv" layer. In this configuration, the kernel size for each conv operation is set to 3×3 to capture more detailed texture features. Therefore, the 15-layer DnCNN contains #482 048 parameters.

E. Gaussian Whitening

To further enhance the denoising capability of the RCI denoising network against non-i.i.d. noise, we also employ GW technique here. GW technique can enhance the model's ability to deal with non-i.i.d noise, which is widely used in many works [21], [32]. The reason behind it is that the noise on different RCIs is normalized to a similar level after GW technique, thereby increasing the model's predictive ability. We also discuss the ablation experiments of the GW technique in the experiment section. Here, we adopt the well-known HySime [60] to estimate the noise covariance Σ. Then the GW of noisy HSI $\mathcal{Z}$ can be expressed as

$$\mathcal{Y} = \mathcal{Z} \times_3 \sqrt{\Sigma^{-1}} \tag{8}$$

where Σ^{-1} denotes its inverse and $\sqrt{\Sigma^{-1}}$ is the square root of Σ^{-1}, resulting $\mathcal{Y}$ is corrupted by Gaussian white noise. Since the noise level of the original HSI tends to be similar after GW, according to the orthogonality invariance of Gaussian noise [37], the noise level of all RCIs also tends to be the same by setting $\boldsymbol{y}_i = \mathcal{Y}(:,:,j)$, $\boldsymbol{w} = \mathbf{V}(:,k)$, $\boldsymbol{u} = \mathcal{U}(:,:,k) = \sum_{i=1}^{n} V(i,k)\mathcal{Y}(:,:,j)$ in Theorem 2. Thus, the μ in (3) can be set to 1. After denoising $\mathcal{Y}$ to get the image $\overline{\mathcal{X}}$, we can use inverse GW on $\overline{\mathcal{X}}$ to get denoised image $\mathcal{X}$, that is,

$$\mathcal{X} = \overline{\mathcal{X}} \times_3 \sqrt{\Sigma}. \tag{9}$$

Theorem 2: Add Gaussian noise $\mathcal{N}(0, \sigma^2)$ independently to n variables $\{\boldsymbol{x}_i\}_{i=1}^n$, resulting in $\{\boldsymbol{y}_i\}_{i=1}^n$. Then, for any

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unit vector $\boldsymbol{w} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_n]$, the noise distribution contained in the mixed variable $\boldsymbol{u} = \sum_i^n \boldsymbol{w}_i \boldsymbol{y}_i$ also follows $e(\boldsymbol{u}) \sim \mathcal{N}(0, \sigma^2)$, where $e(\boldsymbol{u})$ represents the noise contained in variable $\boldsymbol{u}$.

Proof: Firstly, by substituting $\boldsymbol{y}_i = \boldsymbol{x}_i + \boldsymbol{e}_i$ into $\boldsymbol{u} = \sum_i^n \boldsymbol{w}_i \boldsymbol{y}_i$, where $\boldsymbol{e}_i \sim \mathcal{N}(0, \sigma^2)$, then we can obtain

$$\boldsymbol{u} = \sum_i^n \boldsymbol{w}_i \boldsymbol{x}_i + \sum_i^n \boldsymbol{w}_i \boldsymbol{e}_i. \tag{10}$$

Since $\sum_i^n \boldsymbol{w}_i \boldsymbol{x}_i$ contains no noise, according to (10), the noise in $\boldsymbol{u}$ (i.e., $e(\boldsymbol{u})$) is equal to $e(\boldsymbol{u}) = \sum_i^n \boldsymbol{w}_i \boldsymbol{e}_i$. Because $\boldsymbol{e}_i \sim \mathcal{N}(0, \sigma^2)$ and $\boldsymbol{w}$ is a unit vector (i.e., $\sum_{i=1}^n \boldsymbol{w}_i^2 = 1$), we have

$$e(\boldsymbol{u}) \sim \mathcal{N}(0, \sum_{i=1}^n \boldsymbol{w}_i^2 \sigma^2) \sim \mathcal{N}(0, \sigma^2). \tag{11}$$

This completes the proof. ■

F. Algorithm

As our training process for the RCI denoising network is integrated with low-rank decomposition, and to provide a clearer representation of the training process of the RCI denoising network as well as the overall prediction flow of our proposed method, we have consolidated the above process into the following Algorithm 1. For the convenience in description, we abbreviate the entire learning framework as RCI learnable denoiser (i.e., RCI-LD). In Algorithm 1, we set μ in (3) as 1 since the noise level is almost same in all RCIs after GW, the refine parameter as $\lambda = 0.9$, and the total iteration number as $K = 6$. For the 15-layer DnCNN network we chose, we used Kaiming initialization and set the learning rate as 10^{-4}. Furthermore, to capture more characteristics of the tail-end RCIs, the rank of the low-rank decomposition was set to 20 (i.e., $R = 20$) during training. Since our solving algorithm involves a total of six iteration steps, the cumulative parameter count of the network in our model is 2 892 288.

Algorithm 1 RCI-LD

Input: Noisy $\mathcal{Z}$, Iteration Number K, The number of slices of the RCI cube R, Denoiser $\{\Phi_k(\cdot, \Theta)\}_{k=1}^K$.

Initialization: Apply GW on $\mathcal{Z}$ via Eq. (8) to get whitten $\mathcal{Y}$.

while $k \leq K$ **do**
  A). Spectral low-rank decomposition:
  Update $\overline{\mathcal{U}}_k, \mathbf{V}_k$ via Proposition 1.
  B). RCI Deep Denoiser:
  Update $\mathcal{X}_k$ via Eq. (6).
  if Training **then**
    Update Θ in $\Phi_k(\cdot, \Theta)$ via Eq. (7).
  end if
  C). Iteration Regularization:
  Update input noisy data $\mathcal{Y}_{k+1}$ via Eq. (3).
  $k := k + 1$.
end while
Apply IGW on $\mathcal{X}_K$ via Eq. (9) to get recovered image $\hat{X}$.

Output: Recovered HSI Data $\hat{X}$.

IV. Experiments

To demonstrate the effectiveness of our proposed RCI deep denoiser, we selected 14 state-of-the-art denoising methods, including five model-driven methods, namely NNM [15], CTV-RPCA [51], BM4D [60], LLRT [14], and GLF [21], five data-driven methods, namely HSICNN [26], GRN-net [29], QRNN3D [28], T3SC [30], and SMDS [33], and four subspace-based methods, namely RCTV [24], FastHyDe [20], FastHyMix [32], and NGmeet [37]. To ensure fairness, for model-based methods, we adjusted the parameters according to the settings provided in the original papers. As for the deep learning methods, we trained them on the same training dataset as our proposed method. All experiments are conducted on a PC equipped with Intel Core i5-10600KF CPU@4.10 GHz, NVIDIA Tesla P100-PCIE GPU, and 32 GB memory.

A. Simulated Experiments

This experiment aims to evaluate the performance of the proposed network with other competing methods quantitatively. Two clean multispectral image (MSI) datasets, called CAVE[1] and ICVL[2] and two HSI datasets, called DC mall[3] and PaviaU[4] are used in all the synthetic experiments. Specifically, the ICVL dataset contains 201 images, each with a size of $1392 \times 1300 \times 31$; the CAVE dataset contains 31 images, each with a size of $512 \times 512 \times 31$; the DC mall dataset has a size of $200 \times 200 \times 160$; and the PaviaU dataset has a size of $200 \times 200 \times 93$. The large scale of HSI data poses challenges for some fundamental methods, such as LLRT and GLF, making their execution particularly slow. Therefore, in this study, we cropped the ICVL and CAVE datasets, resulting in a final data size of $300 \times 300 \times 31$. In this article, we selected 30 ICVL images and 20 CAVE images as the training data, 1 ICVL image and 1 CAVE image as the validation set, and 10 ICVL images, 10 CAVE images, DC mall, and PaviaU as the test set. The images in the test set are shown in Figs. 6 and 7. For this synthetic experiment, we only need to train one network for all the testing cases. Specifically, we construct a large number of clean-noise data pairs with the same noise level as the test phase. Following the noise settings in [8] and [37], additive Gaussian noise with standard variance σ are added with σ varies from 0.05, 0.1, 0.2, 0.4, $\sigma \in [0, 0.2]$ and $\sigma \in [0, 0.4]$. Three quantitative quality indices are employed: PSNR, SSIM, and SAM, where PSNR and SSIM are the mean value of PSNR and SSIM on each band. The larger the PSNR and SSIM, and the smaller the SAM, the better the recovered image quality. The gray values of the data are re-scaled to [0,1] before conducting experiments.

1) Quantitative Comparison: Firstly, we computed the evaluation metrics for different methods on the ICVL and CAVE HSI datasets under various noise levels, and the results are presented in Table II. From Table II, we can observe the

[1]https://www.cs.columbia.edu/CAVE/databases/multispectral/
[2]http://icvl.cs.bgu.ac.il/hyperspectral/
[3]https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral
[4]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

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Fig. 6. A total of 10 test images in ICVL dataset.



Fig. 7. A total of 10 test images in CAVE dataset.

following phenomena. Firstly, it can be observed that RPCA-based methods do not perform well in removing Gaussian noise. The main reason for this is that RPCA employs the ℓ_1-norm to characterize the noise, which is not effective for Gaussian noise removal. Secondly, the performance of the learning-based methods is generally good, both in terms of denoising performance and denoising time. However, overall, these deep learning methods are still inferior to the proposed RCI-LD in terms of performance. This also indicates the effectiveness of characterizing the hyperspectral low-rank property using low-rank decomposition methods. Thirdly, comparing model-driven and subspace-based methods, we can observe that traditional subspace models achieve comparable performance to the original space models, such as GLF and LLRT. However, their execution time can be significantly reduced, demonstrating the efficiency of subspace-based methods. Fourth, we can observe that using a deep denoiser for RCI, our learning-based approach significantly outperforms FastHyMix, which directly employs pre-trained FFD-net as an RCI denoiser. This result clearly demonstrates the effectiveness of our method. Finally, compared to all other methods, our approach remains superior for both i.i.d and non-i.i.d noise. Particularly for non-i.i.d noise, our method substantially outperforms other subspace-based denoising methods and achieves a 1–1.5 dB higher PSNR value than other deep learning methods.

2) Visual Comparison: To provide a more intuitive demonstration of the restoration performance of all methods, we present the recovered images of some top-performing and representative methods in Figs. 8 and 9. The SPNR values and runtime of each method are also marked below the denoised images. It can be observed that the images restored by our RCI-LD method exhibit less noise residue and better texture information compared to other methods. As shown in Fig. 8, our method can more clearly restore the text in the middle of the image. In Fig. 9, our method effectively removes noise while preserving the image's colors and sharpness of edges. Therefore, it can be concluded that our RCI-LD achieves the best restoration results among all the compared methods.

3) Training Time Comparison: We have already examined the testing times for all methods. Here, we further provide the training times for all deep learning models in this article. Table III lists the training times for all deep learning models on 50 training datasets over 500 epochs. From the table, it can be seen that our method's runtime is at an intermediate level compared to all methods. This phenomenon arises because, during the training phase, in order to capture as much information as possible from the tail-end RCIs, we set the number of RCIs in the low-rank decomposition as $R = 20$. During the testing phase, the rank is generally set to be less than 10, which is the reason for the faster performance of our method.

4) Generalization Experiments: To further demonstrate the generalization of our method, we conducted simulation experiments on two different datasets, namely DC mall and PaviaU, which are distinct from ICVL and CAVE. The corresponding results are presented in Table IV and Fig. 10.

Table IV presents a quantitative evaluation of denoising performance for all competing methods. The table not only reaffirms conclusions from Table II but also reveals additional insights. Firstly, data-driven methods initially lag behind some traditional models such as FastHyde, FastHyMix, GLF, and NGmeet, indicating room for improvement in the generalization ability of deep learning methods. However, given the convenience, efficiency, and reasonable performance during testing, deep learning methods remain highly suitable for practical applications. Secondly, the improvement of our method over NGmeet is not as pronounced as observed in Table II, suggesting a reduction in the generalization performance of RCI-LD. Despite this, our method outperforms NGmeet under i.i.d noise and significantly surpasses other methods under non-i.i.d noise conditions. These findings indicate that our method, relying on the low-rank decomposition framework, possesses a reasonable level of generalization capability. Our network captures the spatial prior of the data without explicitly modeling the low-rank property, aligning well with the strengths of convolutional neural networks.

Furthermore, the restored pseudo-color images and restoration metrics of different comparative methods on the PaviaU dataset were placed in Fig. 10, it is evident that our method significantly outperforms other compared methods in terms of preserving fine details and edge contours. Therefore, the proposed RCI-LD method exhibits significant advantages across multiple dimensions, including denoising runtime, denoising performance, generalization capability, and image fidelity.

B. Real Data Experiments

In this section, we selected three real datasets, ZY1-02D, GF-Shanghai, and Indian Pines, to demonstrate the denoising performance of all competing methods. The introductions for the three datasets are provided below.

The ZY1-02D and GF-Shanghai datasets were captured by the AHSI imager onboard ZiYuan1-02D and Gaofen-5 satellites. ZY1-02D and GF-Shanghai cover rural areas totaling

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS NOISE LEVELS. THE BEST AND SECOND RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Image	σ	Index	Noisy	Model-driven Methods					Data-driven Methods					Subspace-based Methods				
				RPCA	CTV-RPCA	BM4D	LLRT	GLF	HSI-CNN	GRN-net	QR-NN3D	T3SC	SMDS	RCTV	Fast-HyDe	Fast-HyMix	NG-meet	RCI-LD
CAVE	0.05	PSNR	26.02	29.14	35.44	42.39	43.94	43.38	43.36	43.75	43.66	43.58	43.20	39.51	40.68	42.20	43.64	**44.29**
		SSIM	0.392	0.832	0.917	0.975	0.983	0.984	0.982	0.989	0.988	0.986	0.981	0.957	0.969	0.976	0.984	**0.993**
		SAM	0.517	0.311	0.169	0.074	0.047	0.054	0.053	0.048	0.049	0.051	0.075	0.083	0.077	0.055	**0.047**	**0.047**
	0.10	PSNR	19.99	25.88	32.78	38.61	40.76	40.92	40.45	40.87	40.66	40.92	40.14	36.35	37.15	39.14	40.94	**41.24**
		SSIM	0.179	0.674	0.828	0.945	0.972	0.973	0.971	0.977	0.976	0.981	0.955	0.936	0.949	0.968	0.974	**0.984**
		SAM	0.742	0.447	0.259	0.123	0.07	0.075	0.069	0.065	0.067	0.064	0.079	0.118	0.158	0.109	0.063	**0.062**
	0.20	PSNR	13.97	22.43	29.67	34.58	36.91	37.32	37.21	37.63	37.53	37.92	37.44	34.39	34.38	35.64	36.96	**38.34**
		SSIM	0.071	0.482	0.675	0.874	0.945	0.944	0.958	0.964	0.962	0.967	0.963	0.873	0.902	0.906	0.946	**0.971**
		SAM	0.978	0.612	0.379	0.209	0.124	0.109	0.091	0.085	0.087	0.084	0.086	0.134	0.208	0.164	0.106	**0.082**
	0.40	PSNR	7.95	19.12	26.07	30.38	33.48	33.82	33.92	34.26	33.18	34.21	34.02	31.33	30.9	31.82	33.61	**34.67**
		SSIM	0.022	0.321	0.454	0.734	0.905	0.885	0.932	0.939	0.939	0.942	0.936	0.822	0.816	0.806	0.904	**0.944**
		SAM	1.195	0.779	0.529	0.345	0.183	0.172	0.134	0.126	0.129	0.125	0.132	0.278	0.355	0.261	0.158	**0.123**
	0.0-0.20	PSNR	23.12	26.60	33.93	37.74	36.08	41.52	41.96	42.36	42.23	42.54	42.18	36.56	38.32	39.13	29.07	**43.43**
		SSIM	0.314	0.766	0.884	0.914	0.883	0.968	0.963	0.982	0.982	0.986	0.985	0.905	0.946	0.955	0.727	**0.991**
		SAM	0.775	0.410	0.241	0.176	0.269	0.094	0.071	0.057	0.059	0.056	0.062	0.156	0.173	0.106	0.473	**0.051**
	0.0-0.4	PSNR	16.59	23.67	31.51	33.76	30.54	39.42	38.42	39.12	38.89	39.24	38.72	34.03	36.64	36.12	23.93	**40.35**
		SSIM	0.164	0.626	0.779	0.837	0.783	0.968	0.958	0.974	0.972	0.977	0.966	0.887	0.929	0.896	0.524	**0.983**
		SAM	1.025	0.540	0.339	0.289	0.436	0.108	0.112	0.098	0.101	0.095	0.108	0.245	0.238	0.180	0.809	**0.088**
	Time(s)		/	5.16	59.15	143.8	1423	910	3.39	3.25	3.07	4.98	13.12	5.8	4.33	**0.84**	183.9	0.98
ICVL	0.05	PSNR	26.02	32.26	38.94	42.35	44.45	45.23	44.04	44.38	44.23	44.98	45.05	39.77	43.82	44.45	45.44	**45.52**
		SSIM	0.384	0.884	0.944	0.971	0.981	0.983	0.979	0.985	0.983	0.989	0.989	0.963	0.982	0.981	0.984	**0.993**
		SAM	0.568	0.191	0.132	0.095	0.061	0.057	0.062	0.061	0.061	0.058	0.061	0.086	0.082	0.071	**0.055**	0.057
	0.1	PSNR	20.01	28.70	35.56	38.68	41.11	41.66	41.12	41.58	41.34	41.84	41.83	37.62	40.04	41.13	41.91	**42.22**
		SSIM	0.153	0.745	0.879	0.941	0.965	0.967	0.961	0.968	0.971	0.981	0.980	0.936	0.963	0.965	0.971	**0.988**
		SAM	0.806	0.286	0.198	0.139	0.076	0.075	0.081	0.076	0.081	0.077	0.079	0.122	0.113	0.099	**0.071**	0.072
	0.20	PSNR	13.97	24.87	31.86	35.07	37.57	38.24	38.17	38.52	38.29	38.76	38.61	35.28	36.49	37.39	38.32	**39.13**
		SSIM	0.047	0.530	0.748	0.885	0.935	0.941	0.948	0.954	0.951	0.966	0.963	0.893	0.933	0.928	0.943	**0.978**
		SAM	1.051	0.430	0.301	0.199	0.099	0.097	0.102	0.092	0.097	0.092	0.104	0.112	0.159	0.151	0.094	**0.091**
	0.40	PSNR	7.95	21.16	27.65	31.45	34.25	34.97	34.24	34.77	34.48	35.12	34.98	32.53	32.64	33.45	34.56	**35.65**
		SSIM	0.119	0.317	0.514	0.783	0.892	0.918	0.923	0.935	0.931	0.942	0.929	0.864	0.871	0.850	0.897	**0.954**
		SAM	1.25	0.637	0.452	0.296	0.121	0.118	0.128	0.122	0.125	**0.115**	0.142	0.216	0.235	0.235	0.121	0.116
	0.0-0.2	PSNR	22.33	29.74	37.59	37.54	36.11	**45.48**	44.05	44.68	44.72	44.55	44.34	37.35	43.14	41.95	29.95	45.01
		SSIM	0.271	0.844	0.927	0.913	0.832	0.989	0.984	0.988	0.991	0.987	0.988	0.913	0.978	0.972	0.663	**0.993**
		SAM	0.865	0.241	0.161	0.175	0.301	**0.062**	0.085	0.081	0.078	0.082	0.081	0.119	0.102	0.112	0.622	0.072
	0.0-0.4	PSNR	15.67	26.51	34.59	34.16	31.51	41.23	40.13	40.55	40.42	40.82	40.94	33.89	39.07	37.75	26.07	**42.17**
		SSIM	0.125	0.704	0.847	0.843	0.709	0.982	0.981	0.984	0.982	0.986	0.979	0.879	0.957	0.941	0.527	**0.991**
		SAM	1.096	0.347	0.235	0.241	0.391	0.101	0.121	0.108	0.113	0.106	0.117	0.173	0.137	0.179	0.826	**0.097**
	Time(s)		/	5.74	57.45	143.8	1359	880	3.43	3.24	3.02	5.12	12.92	6.5	4.32	**0.94**	181.4	1.2

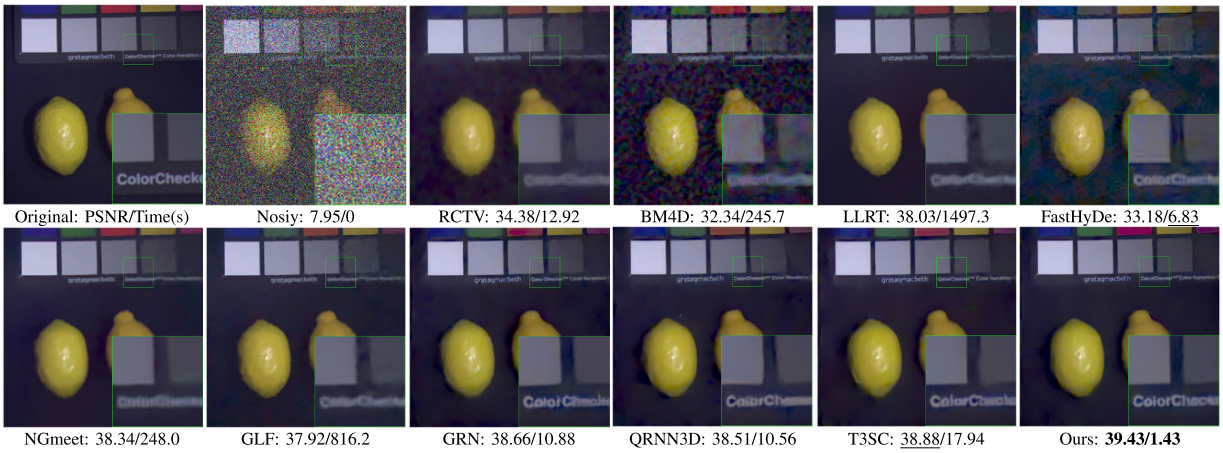


Fig. 8. Recovered false images (R: 27, G: 17, B: 4) on "Lemons" of the CAVE dataset with a size of $512 \times 512 \times 31$ under $\mathcal{N}(0, 0.4^2)$.

more than 200 km^2 in the Tongnan district of Chongqing City, China, and the suburbs of Shanghai City, China. The preprocessed L1-A level data consists of 166 bands and 155 bands, ranging from 400 to 2500 nm, with a ground

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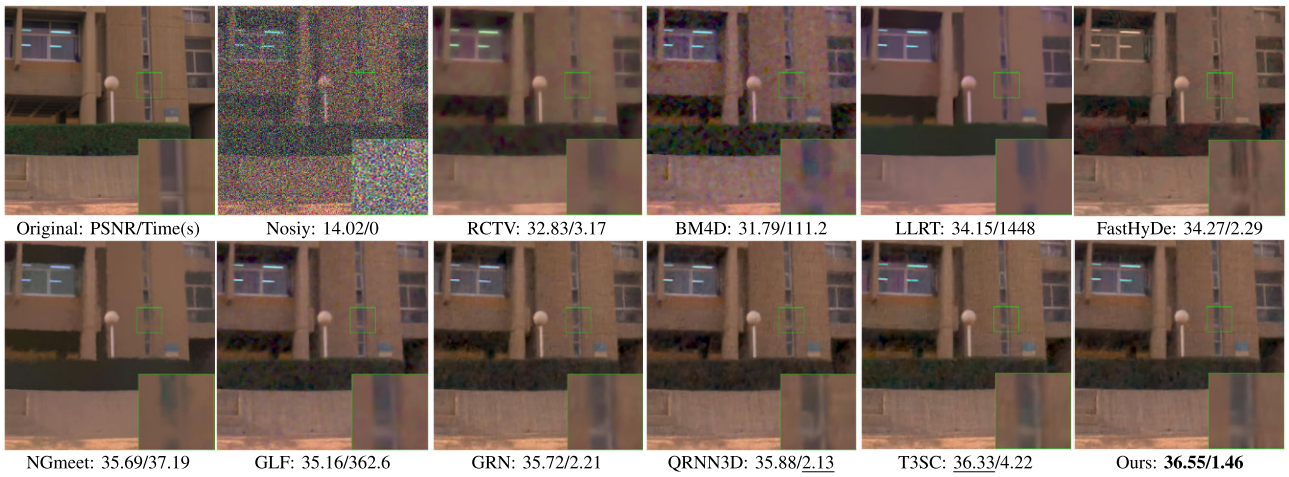


Fig. 9. Recovered false images (R: 27, G: 17, B: 4) on "bguCAMP" of the ICVL dataset with a size of $512 \times 512 \times 31$ under $\mathcal{N}(0, 0.4^2)$. The best and second results are highlighted in bold and underline, respectively.

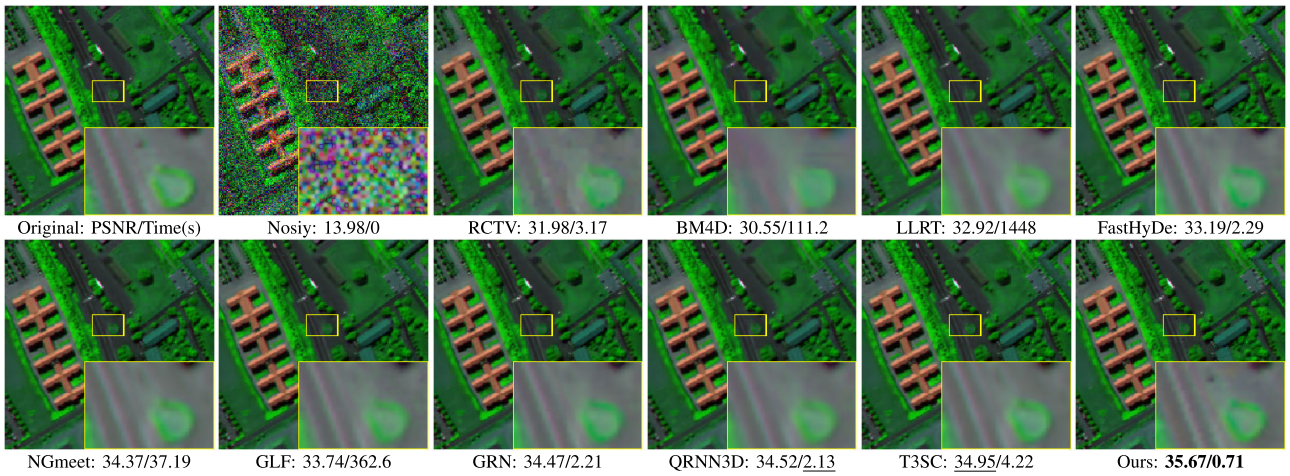


Fig. 10. Recovered images of all competing methods with bands 10-90-50 as R-G-B. The best and second results are highlighted in bold and underline, respectively.

TABLE III

TRAINING TIME FOR ALL COMPETING DEEP LEARNING METHODS WITH 500 EPOCHS UNDER 30 ICVL DATASETS AND 20 CAVE DATASETS ON A V100 GPU

Time	HSICNN	GRN-net	QRNN3D	T3SC	SMDS	Ours
Hours	64.8	34.5	42.7	29.6	32.4	38.2

sampling distance of 30 m. A subset image measuring 300×300 was utilized for method verification. The Indian Pines dataset[5] was imaged by the airborne visible/infrared imaging spectrometer (AVIRIS) in 1992 over a region in the state of Indiana, USA, containing Indian pine trees. The wavelength range of the dataset is 400–2500 nm, with a total of 220 spectral bands. For denoising and classification testing, we typically extract images of size 145×145 pixels from the dataset.

Due to atmospheric interference, photon effects, and physical mechanisms, the three HSI datasets consistently exhibit complex noise contamination, including Gaussian noise, stripe noise, impulse noise, and more. Since these HSIs are real and inherently include noise, obtaining pristine data for numerical calculations is impractical. Hence, we chose to demonstrate the denoising performance of various methods using moderately noisy images. These images not only illustrate the impact of noise on structural integrity but also retain essential features found in clean data, such as texture structure and color fidelity.

For the ZY1-02D dataset, we selected pseudo-color images composed of spectral bands 108, 146, and 166, as well as the spectral curve at the point (250, 150), to showcase the denoising effects of all compared methods in Figs. 11 and 12. For the GF-Shanghai dataset, we selected pseudo-color images composed of spectral bands 108, 146, and 166, as well as the spectral curve at the point (250, 150), to showcase the denoising effects of all compared methods in Figs. 13 and 14. For Indian Pines, we selected pseudo-color images composed of spectral bands 108, 140, and 220, as well as the spectral curve at the point (10, 90), to showcase the denoising effects of all compared methods in Figs. 15 and 16.

Due to the absence of clean images for comparison, we can only judge the denoising quality based on experience. For denoised images, if a method can better preserve the color,

[5]https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html

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TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER VARIOUS NOISE LEVELS ON DC MALL AND PAVIAU. THE BEST AND SECOND RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Image	σ	Index	Noisy	Model-driven Methods					Data-driven Methods					Subspace-based Methods				
				RPCA	CTV-RPCA	BM4D	LLRT	GLF	HSI-CNN	GRN-net	QR-NN3D	T3SC	SMDS	RCTV	Fast-HyDe	Fast-HyMix	NG-meet	RCI-LD
DC mall	0.05	PSNR	26.02	36.02	38.52	36.66	40.35	41.24	39.32	40.77	40.37	40.85	40.29	40.16	40.94	41.39	41.77	**41.86**
		SSIM	0.762	0.983	0.985	0.973	0.991	0.992	0.987	0.990	0.989	0.990	0.984	0.989	0.992	0.992	0.993	**0.995**
		SAM	0.281	0.070	0.061	0.081	0.049	0.045	0.052	0.048	0.052	0.050	0.056	0.051	0.048	0.046	**0.042**	0.045
	0.10	PSNR	20.01	32.07	34.39	32.38	36.42	37.88	36.03	37.22	37.34	37.52	37.25	36.17	37.61	37.67	37.98	**38.18**
		SSIM	0.515	0.959	0.961	0.931	0.978	0.983	0.974	0.981	0.980	0.981	0.978	0.972	0.981	0.981	0.983	**0.986**
		SAM	0.475	0.107	0.093	0.118	0.076	0.067	0.073	0.067	0.068	0.067	0.069	0.075	0.066	0.065	**0.059**	0.063
	0.20	PSNR	13.98	27.99	30.18	28.32	32.34	**34.14**	33.12	32.05	32.63	32.79	32.52	32.36	33.85	33.80	33.81	34.12
		SSIM	0.286	0.901	0.904	0.84	0.946	0.961	0.953	0.962	0.958	0.960	0.962	0.939	0.958	0.957	0.959	**0.965**
		SAM	0.742	0.162	0.141	0.164	0.094	0.084	0.103	0.092	0.096	0.092	0.093	0.089	0.092	0.092	**0.083**	0.086
	0.40	PSNR	7.96	23.91	25.98	24.83	28.48	30.09	29.16	29.78	29.76	29.87	29.55	28.51	29.98	30.04	30.01	**30.66**
		SSIM	0.088	0.782	0.775	0.687	0.871	0.911	0.916	0.933	0.927	0.932	0.930	0.847	0.912	0.908	0.907	**0.942**
		SAM	1.031	0.241	0.217	0.213	0.122	0.116	0.142	0.128	0.132	0.129	0.132	0.183	0.125	0.126	**0.111**	0.118
	0.0-0.20	PSNR	22.51	34.00	37.22	31.02	32.93	42.86	41.85	42.45	42.12	42.36	42.18	40.92	42.65	42.44	32.28	**43.68**
		SSIM	0.547	0.975	0.979	0.904	0.933	0.995	0.989	0.990	0.990	0.992	0.992	0.992	0.994	0.994	0.935	**0.995**
		SAM	0.539	0.086	0.073	0.144	0.127	0.038	0.044	0.039	0.040	0.038	0.039	0.043	0.041	0.041	0.162	**0.036**
	0.0-0.4	PSNR	16.87	29.75	32.95	27.63	30.41	40.34	39.36	40.49	40.15	40.36	40.32	38.65	39.72	38.54	30.27	**41.44**
		SSIM	0.547	0.938	0.945	0.812	0.897	0.991	0.984	0.991	0.991	0.992	0.993	0.986	0.991	0.985	0.909	**0.994**
		SAM	0.786	0.139	0.115	0.178	0.141	0.049	0.054	0.051	0.052	0.052	0.051	0.058	0.053	0.065	0.198	**0.048**
	Time(s)		/	13.16	145.2	186.3	1475	304.2	2.75	2.61	2.52	4.86	10.52	5.82	3.21	**0.44**	39.42	0.93
PaviaU	0.05	PSNR	26.02	35.19	37.86	38.36	40.9	41.54	40.23	40.87	40.89	41.27	41.03	40.51	40.79	40.87	42.27	**42.66**
		SSIM	0.632	0.957	0.965	0.966	0.982	0.984	0.972	0.975	0.974	0.985	0.983	0.978	0.982	0.981	0.986	**0.994**
		SAM	0.255	0.064	0.054	0.057	0.044	0.038	0.051	0.045	0.046	0.041	0.042	0.046	0.042	0.042	**0.035**	0.037
	0.1	PSNR	19.99	30.45	34.03	34.46	36.88	37.82	37.04	37.36	37.25	38.12	38.03	36.44	37.13	37.18	38.57	**39.41**
		SSIM	0.365	0.889	0.919	0.927	0.959	0.967	0.962	0.966	0.965	0.967	0.969	0.957	0.964	0.960	0.971	0.991
		SAM	0.457	0.101	0.080	0.081	0.068	0.051	0.061	0.057	0.059	0.052	0.053	0.073	0.058	0.058	0.047	**0.046**
	0.20	PSNR	13.98	25.78	30.18	30.52	33.02	34.08	33.57	34.05	33.95	34.32	34.11	32.46	33.63	33.60	34.78	**35.33**
		SSIM	0.157	0.752	0.825	0.845	0.918	0.932	0.928	0.939	0.941	0.950	0.948	0.892	0.924	0.923	0.941	**0.976**
		SAM	0.747	0.158	0.122	0.114	0.086	0.071	0.084	0.077	0.072	0.071	0.072	0.087	0.079	0.079	0.066	**0.065**
	0.40	PSNR	7.96	21.52	26.34	27.05	29.51	30.46	30.13	30.55	30.78	30.88	30.62	29.69	30.43	30.31	31.19	**31.61**
		SSIM	0.049	0.549	0.664	0.725	0.837	0.861	0.883	0.904	0.901	0.907	0.903	0.842	0.861	0.859	0.881	**0.937**
		SAM	1.057	0.241	0.190	0.157	0.111	0.095	0.114	0.103	0.106	0.102	0.101	0.112	0.102	0.104	**0.088**	0.089
	0.0-0.2	PSNR	23.45	32.45	36.89	33.38	31.12	**44.08**	42.42	42.96	42.88	43.09	43.14	42.27	43.02	41.96	32.79	43.68
		SSIM	0.478	0.943	0.957	0.904	0.812	0.991	0.984	0.987	0.987	0.989	0.990	0.987	0.989	0.985	0.863	**0.994**
		SAM	0.501	0.085	0.060	0.106	0.156	**0.032**	0.045	0.041	0.042	0.041	0.040	0.045	0.037	0.039	0.189	**0.032**
	0.0-0.4	PSNR	16.74	28.21	33.43	29.62	26.74	39.93	39.07	39.58	39.23	39.45	39.27	38.38	39.24	39.30	29.38	**40.41**
		SSIM	0.274	0.876	0.912	0.814	0.646	0.981	0.975	0.981	0.977	0.979	0.977	0.973	0.977	0.978	0.813	**0.991**
		SAM	0.813	0.133	0.094	0.145	0.242	0.047	0.058	0.052	0.056	0.055	0.056	0.059	0.052	0.053	0.205	**0.047**
	Time(s)		/	7.5668	102.760	111.15	1448	362.56	2.38	2.21	2.13	4.22	9.12	3.17	2.29	**0.51**	37.19	0.71

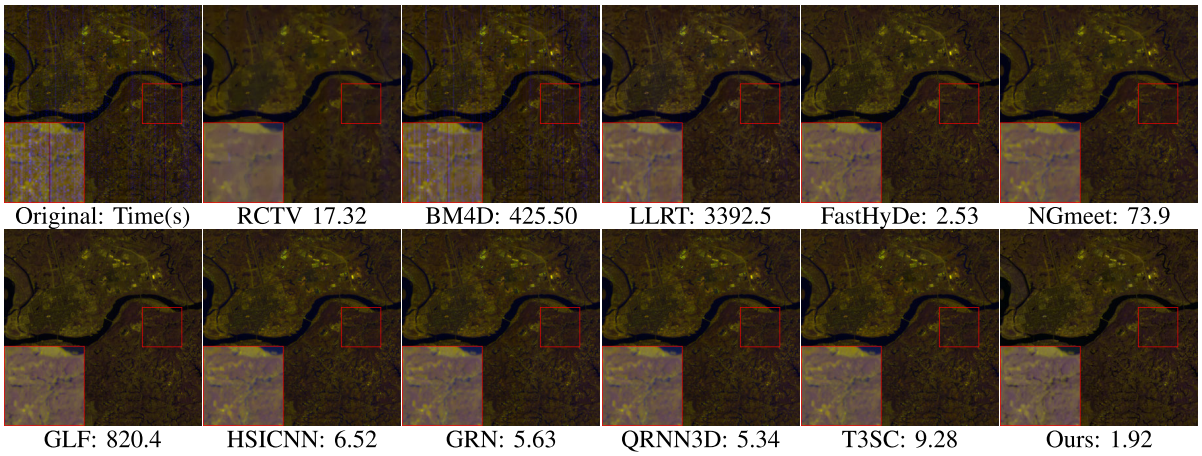


Fig. 11. Recovered images of all competing methods with bands 108-146-166 as R-G-B on ZY1-02D dataset.

texture, and local smoothness of the image, and remove more noise, it can be considered to achieve better denoising results. Regarding the spectral curves, if a method can denoise the spectral curve while effectively preserving the trend of the

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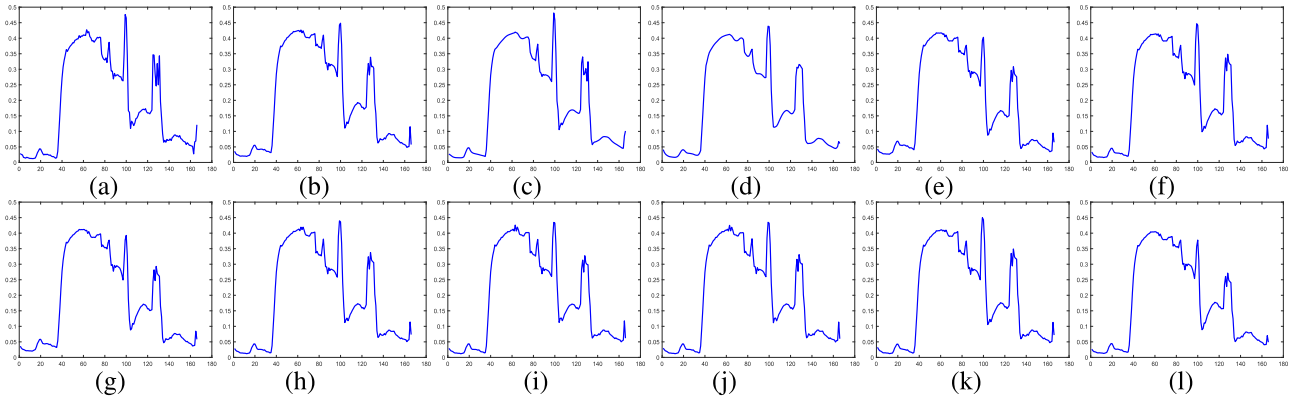


Fig. 12. Denoised spectral signatures of point (250, 150) on real ZY1-02D dataset by different methods. The x-axis of each subgraph represents the order of spectral bands, and the y-axis represents the grayscale values. (a) Noisy. (b) RCTV. (c) BM4D. (d) LLRT. (e) FastHyDe. (f) NGmeet. (g) GLF. (h) HSICNN. (i) GRN. (j) QRNN3D. (k)T3SC. (l) Ours.

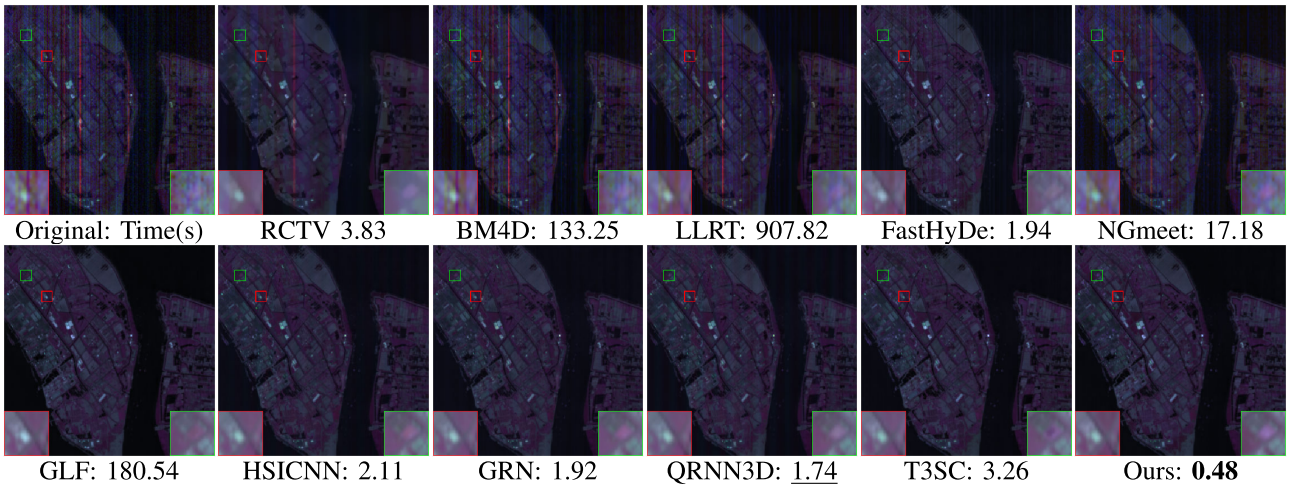


Fig. 13. Recovered images of all competing methods with bands 42-152-155 as R-G-B on GF-Shanghai dataset.

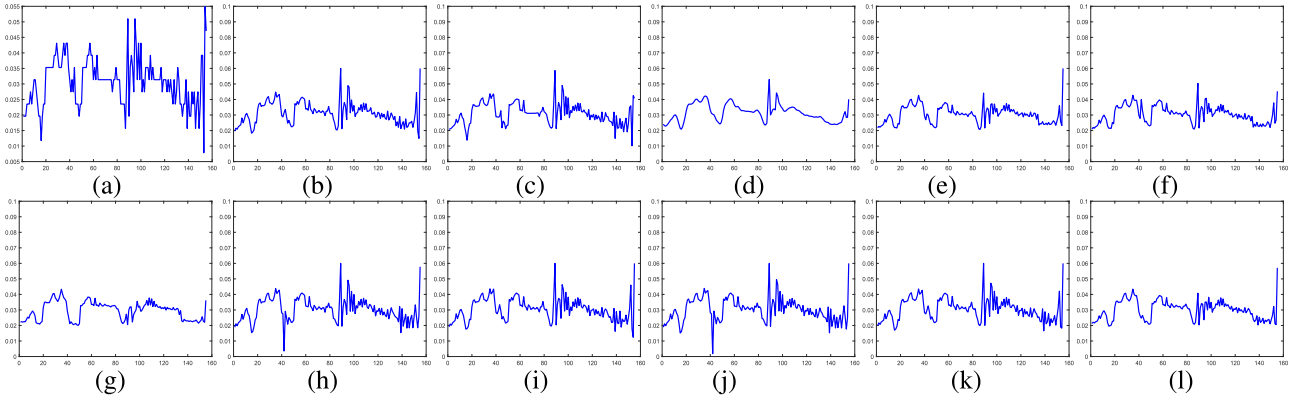


Fig. 14. Restored spectral signatures of point (200, 200) on real GF-Shanghai dataset by different methods. (a) Noisy. (b) RCTV. (c) BM4D. (d) LLRT. (e) FastHyDe. (f) NGmeet. (g) GLF. (h) HSICNN. (i) GRN. (j) QRNN3D. (k)T3SC. (l) Ours.

original noisy HSI spectral curve (such as local smoothness) and still maintaining the characteristics of a specific point, then the method can be considered effective for denoising. Combining the denoised images and spectral curves, it is evident that our method has more discriminative features in terms of color fidelity, noise removal, and structural preservation. Additionally, we placed the restoration time of different methods under each restored subfigure, and from this value, we can see the efficiency of the proposed method.

HSI denoising is essential to enhance downstream tasks. To assess denoising performance, we conduct a classification experiment on the Indian Pines dataset before and after denoising, examining the impact of different denoising processes on downstream applications. We employ support

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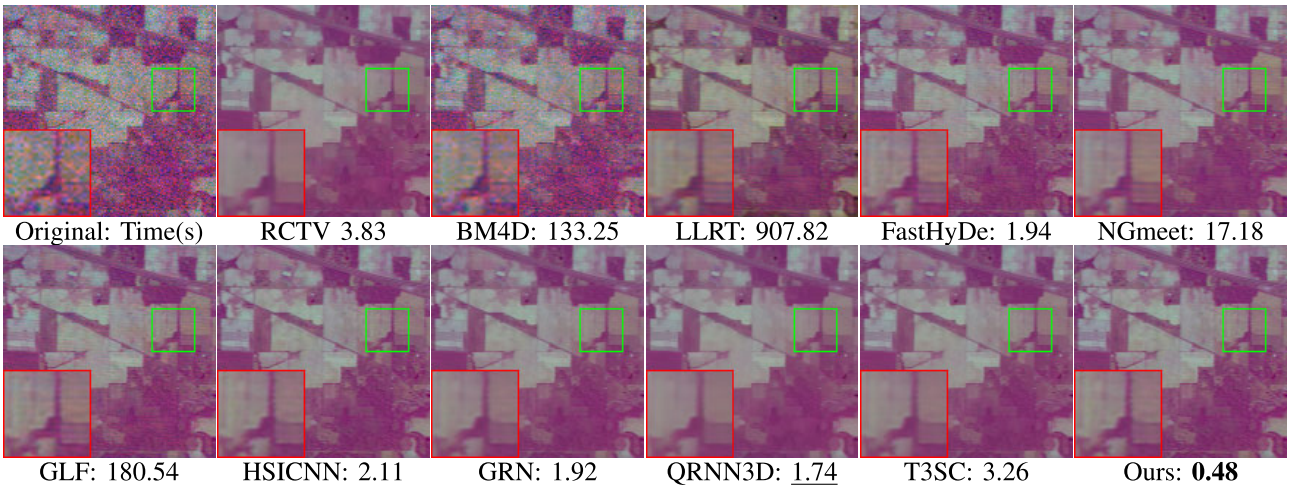


Fig. 15. Recovered images of all competing methods with bands 108-140-220 as R-G-B on Indian Pines dataset.

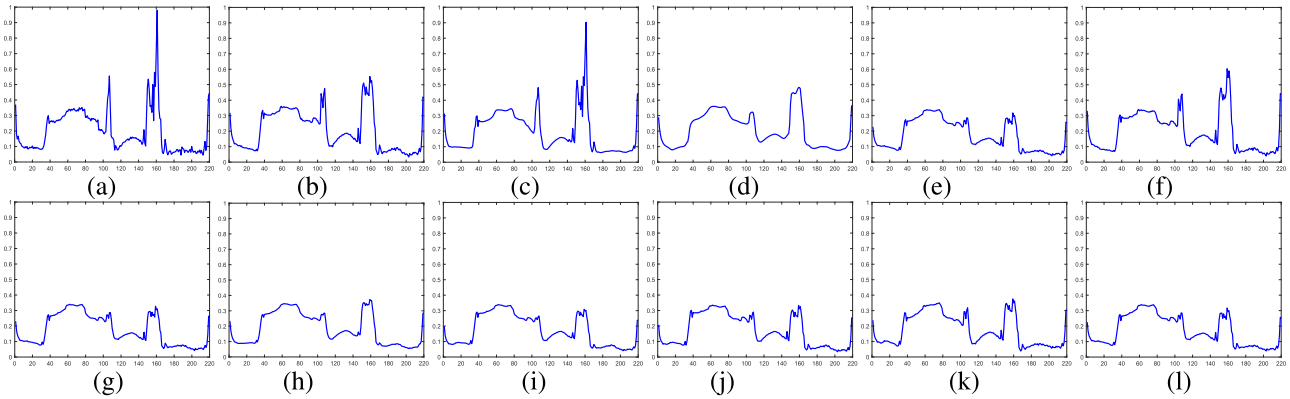


Fig. 16. Restored spectral signatures of point (10, 90) on real Indian Pines dataset by different methods. (a) Noisy. (b) RCTV. (c) BM4D. (d) LLRT. (e) FastHyDe. (f) NGmeet. (g) GLF. (h) HSICNN. (i) GRN. (j) QRNN3D. (k) T3SC. (l) Ours.

vector machine (SVM) classification using a ground truth label with 16 categories, ranging from 20 to 2455 samples per category. Ten samples from each category are randomly chosen for training, and the rest for testing. Three quantitative assessments—overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa)—are utilized to evaluate the classification results [61].

For a fair comparison, we repeat the classification experiment on 10 randomly selected training and testing datasets, recording the best classification results based on AA in Table V. It is evident from Table V that all comparison methods enhance classification performance, with our method yielding the best values across all metrics. Additionally, the label maps of the classification results are shown in Fig. 17. The label map on the denoised data by our method closely aligns with the true map, consistent with the quantitative evaluation. This outcome further confirms the superior performance of our denoising method over other comparison methods.

C. Ablation Experiments

Next, we will sequentially demonstrate the impact of the network structure, GW technology, the rank R in low-rank decomposition, the number of iterations steps K, the trade-off parameter λ, and training mode.

1) Influence of Network Structure: In Section III-D, we analyzed the rationale behind selecting DnCNN as the RCI denoiser. Specifically, the scale differences in all RCIs are substantial, and the noise distribution in all RCIs tends to be the same after GW. Here we further show the impact of different network structures on denoising performance. The results of different RCI network structures are listed in Table VI, where "N" and "GW" represent the abbreviation of scale normalization and GW, respectively. It can be easily observed that the DnCNN network is a good choice for the RCI denoiser.

2) Influence of GW: Comparing the PSNR values of DnCNN with GW and without GW in Table VI, it can be found that the role of GW is mainly to improve the ability of the model to deal with non-i.i.d noise. We have analyzed the reason behind it in detail in Section III-E.

3) Influence of Rank: In a low-rank decomposition framework, a small rank R would cause the loss of information of HSI, while the big rank will bring in the noise and increase the denoising time. However, surprisingly, our method is not sensitive to rank. Fig. 18 shows the influence of different ranks on the restoration results under different noise levels. Combining the PSNR and SSIM curves, it can be observed that, although the optimal performance is achieved at some

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TABLE V
CLASSIFICATION ACCURACY COMPARISON ON AVIRIS INDIAN PINES DATA. THE BEST RESULT IS HIGHLIGHTED IN BOLD

	Noisy HSI	RCTV	BM4D	LLRT	FastHyDe	NGmeet	GLF	HSICNN	GRN	QRN	T3SC	Ours
OA	0.4994	0.7411	0.5640	0.6832	0.5867	0.6034	0.6027	0.6749	0.7507	0.7543	0.7597	0.7630
AA	0.5976	0.8261	0.6737	0.7837	0.7216	0.7004	0.7391	0.7839	0.8340	0.8347	0.8388	0.8492
kappa	0.4390	0.7073	0.5120	0.6428	0.5407	0.5547	0.5588	0.6350	0.7182	0.7220	0.7282	0.7323

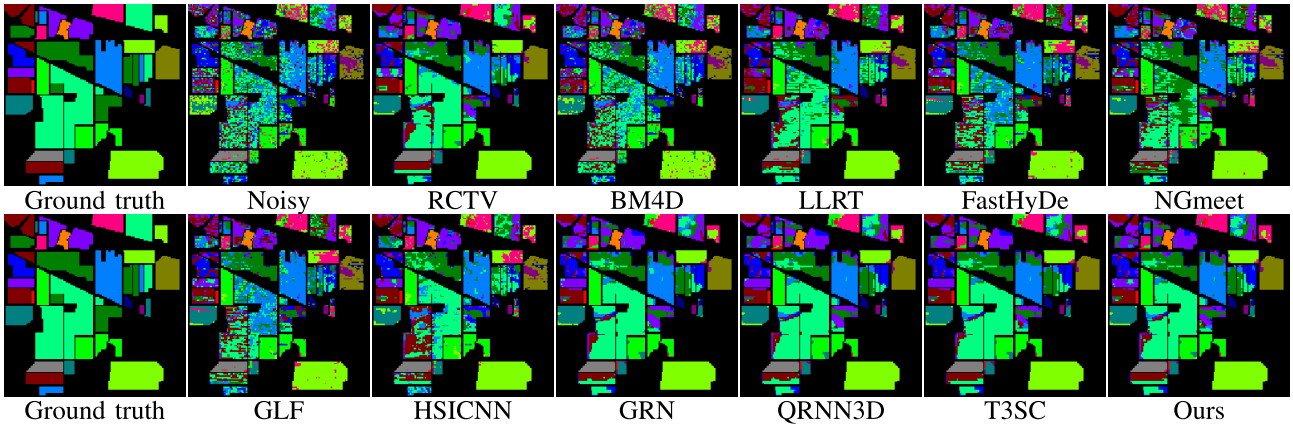


Fig. 17. Classification results by performing SVM on Indian Pines dataset before and after denoising.

TABLE VI
PSNR COMPARISON OF CAVE DATASET FOR DIFFERENT RCI DENOISERS UNDER GAUSSIAN NOISE WITH $\mathcal{N}(0, \sigma^2)$

RCI denoiser	CNN	CNN	FFDnet	DnCNN	DnCNN	DnCNN
Technique	GW	GW+N	GW	GW	GW+N	/
$\sigma = 0.10$	17.28	39.46	40.78	42.22	42.19	**42.32**
$\sigma \in [0, 0.4]$	14.69	39.52	40.64	42.37	**42.44**	35.93

intermediate rank, the performance of our method does not decrease significantly with the enhancement of rank. This is mainly because the later RCI images correspond to less important principal components, and our method learns a denoiser for all RCIs, ensuring that introducing later RCI images does not have much side effect on the results. Fig. 19 illustrates a positive correlation between denoising time and rank. Therefore, considering overall performance and denoising time, it is advisable to choose a moderate rank.

4) Influence of Iteration Number K: LLRT [14] and NGmeet [37] have shown that iterations can enhance denoising performance robustness. However, excessive iterations can be time-consuming, requiring the selection of an optimal iteration value to balance denoising time and performance. In Fig. 20, it is evident that our RCI-LD achieves stable performance with only three steps, and the metric indices consistently surpass those of other competing methods at each stage.

5) Influence of Trade-off Parameter λ: λ is a crucial hyperparameter since it determines the ratio between the original noisy image and the clean image obtained from the previous restoration step in the next network input. However, since our model is trained with a fixed λ, the model will learn parameters that are suitable for that λ. As a result, the testing performance of the model remains fairly consistent across fixed lambdas, as observed in Table VII. Additionally, it can be seen from

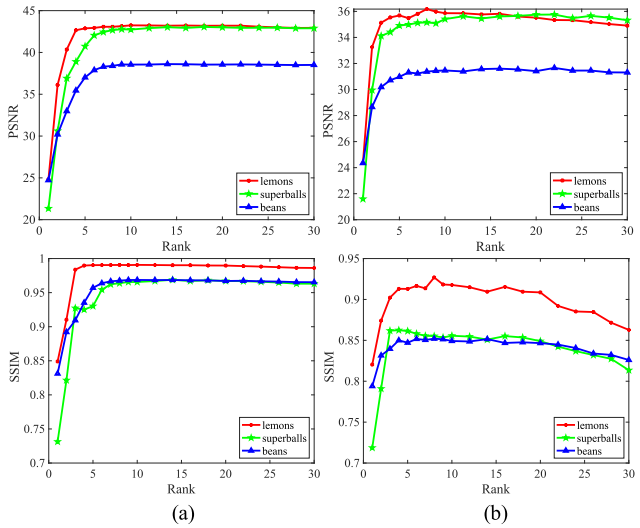


Fig. 18. Relationship between rank and restoration performance under different types of noise and diverse datasets. (a) Std = 0.10. (b) Std = 0.40.

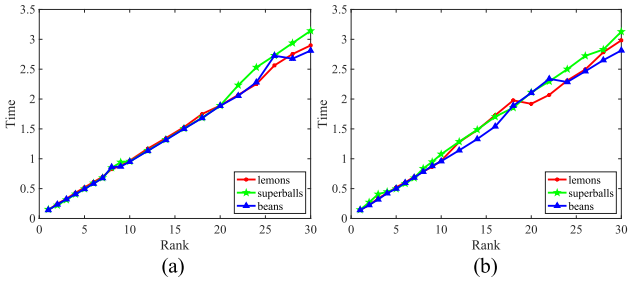


Fig. 19. Relationship between rank and runtime under different types of noise and diverse datasets. (a) Std = 0.10. (b) Std = 0.40.

Table VII that the performance is slightly better when $\lambda = 0.9$. Therefore, in this article, we choose $\lambda = 0.9$. Additionally, from Table VIII, it can be seen that the testing λ is best when

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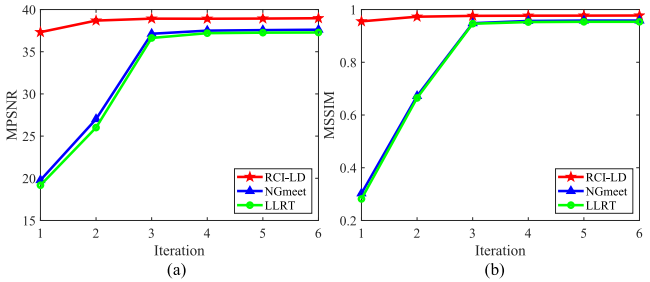


Fig. 20. Relationship between the denoising performance/time and iteration under $\mathcal{N}(0, 0.4^2)$. (a) MPSNR versus iteration. (b) MSSIM versus iteration.

TABLE VII

DISPLAY OF THE RESTORATION PERFORMANCE ON THE DC MALL DATASET FOR MODELS TRAINED UNDER DIFFERENT FIXED VALUES OF λ. THE SAME λ IS USED DURING BOTH TRAINING AND TESTING

Std	$\lambda = 0.9$		$\lambda = 0.7$		$\lambda = 0.5$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
0.05	41.86	**0.9951**	**41.88**	**0.9951**	41.87	0.9949
0.10	38.18	0.9859	**38.29**	0.9863	38.22	**0.9864**
0.20	**34.12**	**0.9657**	34.04	0.9626	33.76	0.9585
0.40	**30.66**	**0.9426**	30.35	0.9242	30.14	0.9229

TABLE VIII

DISPLAY OF THE TEST RESULTS OF THE MODEL OBTAINED BY TRAINING $\lambda = 0.9$ ON DC MALL UNDER DIFFERENT λ

Std	Metric	λ in test mode					
		1	0.9	0.7	0.5	0.3	0.1
0.05	PSNR	41.72	41.86	41.35	40.69	40.48	40.34
	SSIM	0.9949	0.9951	0.9929	0.9910	0.9902	0.9898
0.1	PSNR	38.13	38.18	36.73	35.75	35.45	35.25
	SSIM	0.9859	0.9859	0.9747	0.9659	0.9623	0.9601
0.2	PSNR	30.43	34.12	31.58	27.31	26.62	26.42
	SSIM	0.9681	0.9657	0.9192	0.8931	0.8800	0.8746
0.4	PSNR	30.55	30.66	27.14	25.87	24.94	24.81
	SSIM	0.9335	0.9426	0.8134	0.7478	0.7091	0.7073

same with the training λ. The farther away the testing and training λ are, the worse the performance is.

6) Influence of Training Mode: RCI-LD is to embed RCI deep denoiser into low-rank decomposition for training. There are two similar training ways, one is to train deep denoiser on the original image and then embed the denoiser into the LRTD (i.e., LR-DD in Table IX), and the other is to directly train an RCI denoiser and then embed denoiser into the LRTD (i.e., RCI-LD* in Table IX). For directly training an RCI denoiser, we take the following steps: 1) perform spectral low-rank decomposition on the HSI data to obtain the RCIs; 2) train the obtained RCIs to get a DnCNN denoiser; and 3) put it back into the LRTD for denoising. From Table IX, we can observe that the performance of LR-DD is even inferior to DnCNN. The reason behind this is that LR-DD can be considered as a low-rank decomposition on top of DnCNN, leading to information loss. On the other hand, the performance of RCI-LD* is not as good as the proposed RCI-LD in this article. This is because the noise is added to the original HSI data instead of the RCIs. Therefore, the network trained on pairs of noisy-clean RCI data does not perform as well as our method, which can better fit the noise distribution in the original HSI data.

TABLE IX

QUANTITATIVE COMPARISON OF CAVE DATASET FOR DIFFERENT TRAINING MODE UNDER $\mathcal{N}(0, 0.1^2)$. THE BEST RESULTS ARE BOLDED

Training mode	DnCNN	RCI-LD	RCI-LD*	LR-DD
PSNR	31.79	**42.35**	38.68	27.68
SSIM	0.9204	**0.9864**	0.9687	0.7643
TIMES	2.08	**1.26**	1.26	12.45

V. CONCLUSION

In this article, we introduce a novel subspace-based approach to HSI denoising using pairs of noisy and clean HSI images. Instead of relying on clean RCIs, we update the parameters in the RCI denoiser by parametrically representing the restored HSI. This enables us to learn a specialized deep RCI denoiser, eliminating the need for ground truth clean RCIs. Furthermore, we propose a unified deep denoiser that can handle all RCIs by leveraging residual learning and GW techniques. This ensures that our model is capable of effectively denoising HSIs with various noise levels and ranks, making it highly versatile. Extensive results demonstrate the superiority of our model in both denoising performance and runtime compared to traditional methods and other deep learning-based approaches.

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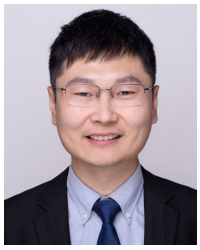
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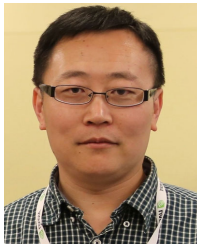
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